


## *DEPARTMENT OF SOFTWARE ENGINEERING*

# CLOUD COMPUTING

**ASSIGNMENT NO 01:**

## Git, Gitea, Git LFS, and GitHub Pages

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**Course:** Cloud Computing (BSE-410)

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# Introduction

This assignment demonstrates the use of Git and remote repositories with Gitea and GitHub. It also covers Git Large File Storage (Git LFS) for tracking large files and GitHub Pages for publishing a portfolio website.

The practical work was performed using an Ubuntu server, Git, Docker Compose, Gitea, GitHub, and a Windows PC for preparing the final submission.

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# Task 1: Gitea

## Objective

The objective of this task was to run Gitea on the Ubuntu server, create an `Assignment01` repository, and push the local Git repository to it.

## Steps Performed

1. Connected to the Ubuntu server and checked the installed Git, Git LFS, and Docker tools.

Cloned the provided Gitea setup repository and entered its directory:
git clone https://github.com/WaqasSaleem97/Gitea.git

2. cd Gitea

Started the Gitea services with Docker Compose:
docker compose up -d

3. docker compose ps
4. Opened Gitea in a browser at:
   **http://192.168.121.129:3000**
5. Completed the initial configuration, created the Gitea account, and created a repository named `Assignment01`.

In the local Ubuntu repository, added Gitea as a remote and pushed the `main` branch:
git remote add gitea
http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git

6. git push -u gitea main

## Result

The Gitea containers started successfully. The `Assignment01` repository was created, and the local `main` branch was pushed to Gitea.

**Gitea repository:**
http://192.168.121.129:3000/samreenzia75/Assignment01

## Evidence

### Screenshot 1: Gitea running

`task1_gitea_running.png.png`



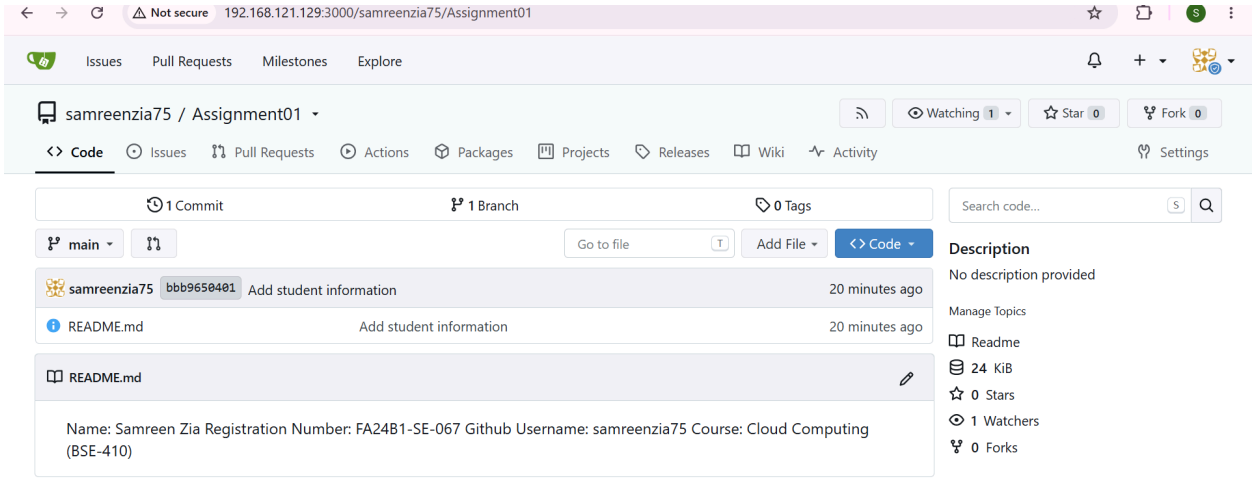

### Screenshot 2: Successful push to Gitea

`task1_gitea_push.png.png`

```
samreenzia75@ubuntu:~/Assignment01$ git config user.email "samreenziaahmed@gmail.com"
samreenzia75@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) bbb9650] Add student information
 1 file changed, 4 insertions(+)
 create mode 100644 README.md
samreenzia75@ubuntu:~/Assignment01$ git remote add gitea http://samreenzia75@127.0.0.1.3000/samreenzia75/Assignment01.git
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1.3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1.3000/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
fatal: unable to access 'http://127.0.0.1.3000/samreenzia75/Assignment01.git/': Could not resolve host: 127.0.0.1.3000
samreenzia75@ubuntu:~/Assignment01$ git remote add gitea http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git
error: remote gitea already exists.
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
fatal: unable to access 'http://127.0.0.1.3000/samreenzia75/Assignment01.git/': Could not resolve host: 127.0.0.1.3000
samreenzia75@ubuntu:~/Assignment01$ git remote add gitea http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git
error: remote gitea already exists.
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1.3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1.3000/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
fatal: unable to access 'http://127.0.0.1.3000/samreenzia75/Assignment01.git/': Could not resolve host: 127.0.0.1.3000
samreenzia75@ubuntu:~/Assignment01$ git remote set-url gitea http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/samreenzia75/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
samreenzia75@ubuntu:~/Assignment01$ _
```

## Screenshot 3: Gitea Assignment01 repository

task1_gitea_repository.png.png



# Task 2: GitHub

## Objective

The objective of this task was to connect the same local repository to GitHub and push the `main` branch.

### Steps Performed

1. Created a public GitHub repository named `Assignment01`.
2. Added the GitHub repository as a second remote:
   git remote add github https://github.com/samreenzia75/Assignment01.git
3. Checked that both Gitea and GitHub remotes were configured:
   git remote -v
4. Pushed the `main` branch to GitHub:
   git push -u github main

## Result

The local repository was connected to both Gitea and GitHub. The `main` branch was pushed successfully to GitHub.

**GitHub repository:**
https://github.com/samreenzia75/Assignment01

## Evidence

### Screenshot 1: Configured Git remotes

`task2_remotes.png.png`

```
fatal: unable to access 'http://127.0.0.1.3000/samreenzia75/Assignment01.git/': Could not resolve host: 127.0.0.1.3000
samreenzia75@ubuntu:~/Assignment01$ git remote set-url gitea http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://samreenzia75@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 330.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/samreenzia75/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
samreenzia75@ubuntu:~/Assignment01$ cd Assignment01
-bash: cd: Assignment01: No such file or directory
samreenzia75@ubuntu:~/Assignment01$ cd~ /Assignment01
Command 'cd~' not found, did you mean:
  command 'cde' from deb cde (0.1+git9-g551e54d-3)
  command 'cdv' from deb erlang-observer (1:27.3.4.6+dfsg-1)
  command 'cdi' from deb cdo (2.5.4-1)
  command 'cdw' from deb cdw (0.8.1-3build4)
  command 'cdp' from deb irpas (0.10-10build1)
  command 'cdo' from deb cdo (2.5.4-1)
  command 'cdb' from deb tinycdb (0.81-2build1)
  command 'cd5' from deb cd5 (0.1-4build1)
Try: sudo apt install <deb name>
samreenzia75@ubuntu:~/Assignment01$
samreenzia75@ubuntu:~/Assignment01$
samreenzia75@ubuntu:~/Assignment01$ git remote add github https://github.com/samreenzia75/Assignment01.git
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (push)
github  https://github.com/samreenzia75/Assignment01.git (fetch)
github  https://github.com/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ _
```

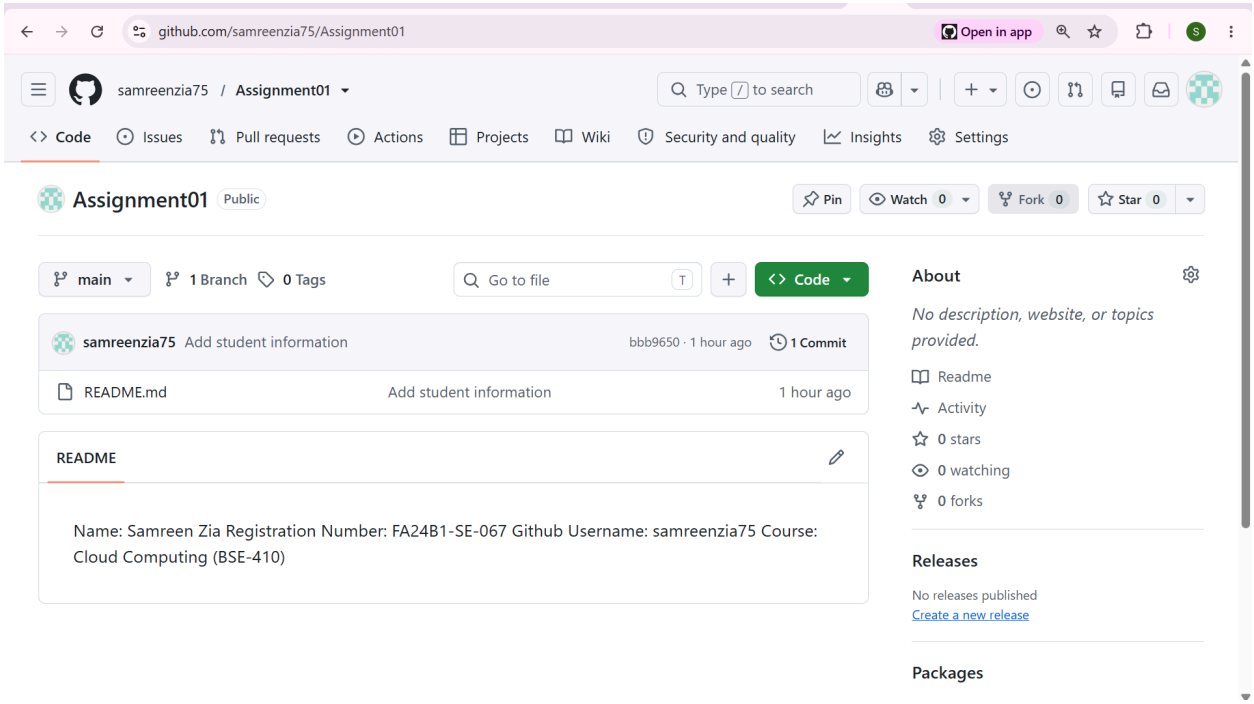
## Screenshot 2: Successful push to GitHub

task2_github_push.png.png

```
  command 'cde' from deb cde (0.1+git9-g551e54d-3)
  command 'cdv' from deb erlang-observer (1:27.3.4.6+dfsg-1)
  command 'cdi' from deb cdo (2.5.4-1)
  command 'cdw' from deb cdw (0.8.1-3build4)
  command 'cdp' from deb irpas (0.10-10build1)
  command 'cdo' from deb cdo (2.5.4-1)
  command 'cdb' from deb tinycdb (0.81-2build1)
  command 'cd5' from deb cd5 (0.1-4build1)
Try: sudo apt install <deb name>
samreenzia75@ubuntu:~/Assignment01$
samreenzia75@ubuntu:~/Assignment01$
samreenzia75@ubuntu:~/Assignment01$ git remote add github https://github.com/samreenzia75/Assignment01.git
samreenzia75@ubuntu:~/Assignment01$ git remote -v
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (fetch)
gitea   http://samreenzia75@127.0.0.1:3000/samreenzia75/Assignment01.git (push)
github  https://github.com/samreenzia75/Assignment01.git (fetch)
github  https://github.com/samreenzia75/Assignment01.git (push)
samreenzia75@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': samreenzia75
Password for 'https://samreenzia75@github.com':
samreenzia75@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': samreenzia75
Password for 'https://samreenzia75@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': samreenzia75
Password for 'https://samreenzia75@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': SAMREENZIA75
Password for 'https://SAMREENZIA75@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/samreenzia75/Assignment01.git/'
samreenzia75@ubuntu:~/Assignment01$ git remote add github https://github.com/samreenzia75/Assignment01.git
error: remote github already exists.
samreenzia75@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': samreenzia75
Password for 'https://samreenzia75@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 330 bytes | 165.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/samreenzia75/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
samreenzia75@ubuntu:~/Assignment01$
```

## Screenshot 3: GitHub Assignment01 repository

task2_github_repository.png.png



# Task 3: Git LFS

## Objective

The objective of this task was to configure Git LFS, create three files larger than 100 MiB, track them with Git LFS, and push them to GitHub.

## Steps Performed

Initialized Git LFS in the local repository and configured it to track `.bin` files:
git lfs install
git lfs track "*.bin"

1. cat .gitattributes

Created three files larger than 100 MiB:
truncate -s 101M large-file-1.bin
truncate -s 102M large-file-2.bin

2. truncate -s 103M large-file-3.bin

Added the LFS rule and files, then checked the tracked files and sizes:
git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin

3. git lfs ls-files --size

Committed the files and pushed them to GitHub:
git commit -m "Add three files using Git LFS"

4. git push github main

## Result:

Git LFS tracked all three `.bin` files. The files were approximately 101 MiB, 102 MiB, and 103 MiB. All three LFS objects and the commit were pushed successfully to GitHub.

| File | Size created |
| --- | --- |
| `large-file-1.bin` | 101 MiB |
| `large-file-2.bin` | 102 MiB |

| | |
|---|---|
| `large-file-3.bin` | 103 MiB |

# Evidence

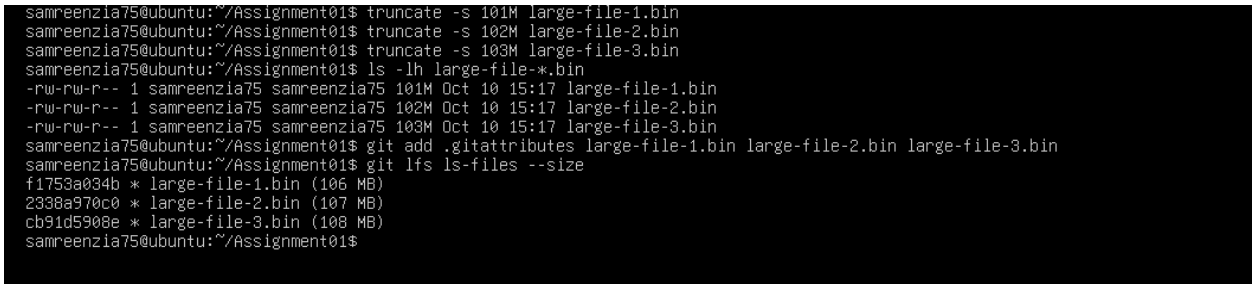
### Screenshot 1: Git LFS setup and tracking rule

`task3_lfs_setup.png.png`



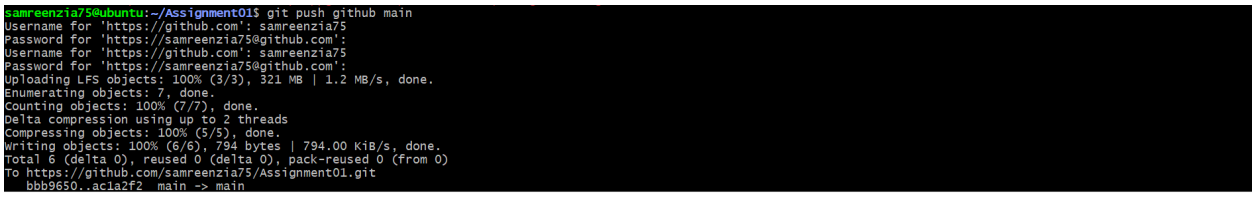

### Screenshot 2: LFS-tracked files and sizes

`task3_lfs_files.png.png`




### Screenshot 3: Successful GitHub LFS push

`task3_lfs_push.png.png`




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# Task 4: GitHub Pages

## Objective

The objective of this task was to create a portfolio homepage and publish it using GitHub Pages.

## Steps Performed

1. Created a public GitHub repository named `samreenzia75.github.io`.
2. Created an `index.html` file containing the portfolio homepage.
3. Opened the repository's **Settings** → **Pages** section.
4. Selected **Deploy from a branch**, chose the `main` branch and the `/(root)` folder, and saved the settings.
5. Opened the published website and confirmed that the portfolio page was live.

## Result

The portfolio repository was published using GitHub Pages.

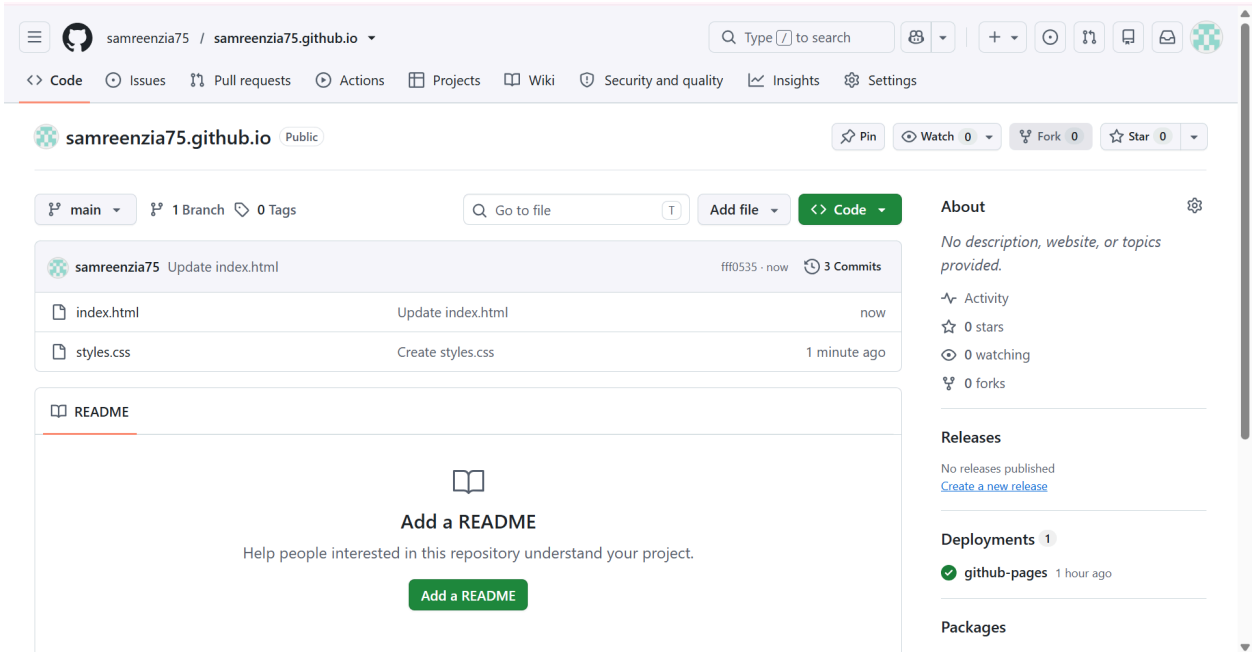
**Portfolio repository:**
https://github.com/samreenzia75/samreenzia75.github.io

**Published website:**
https://samreenzia75.github.io/

## Evidence

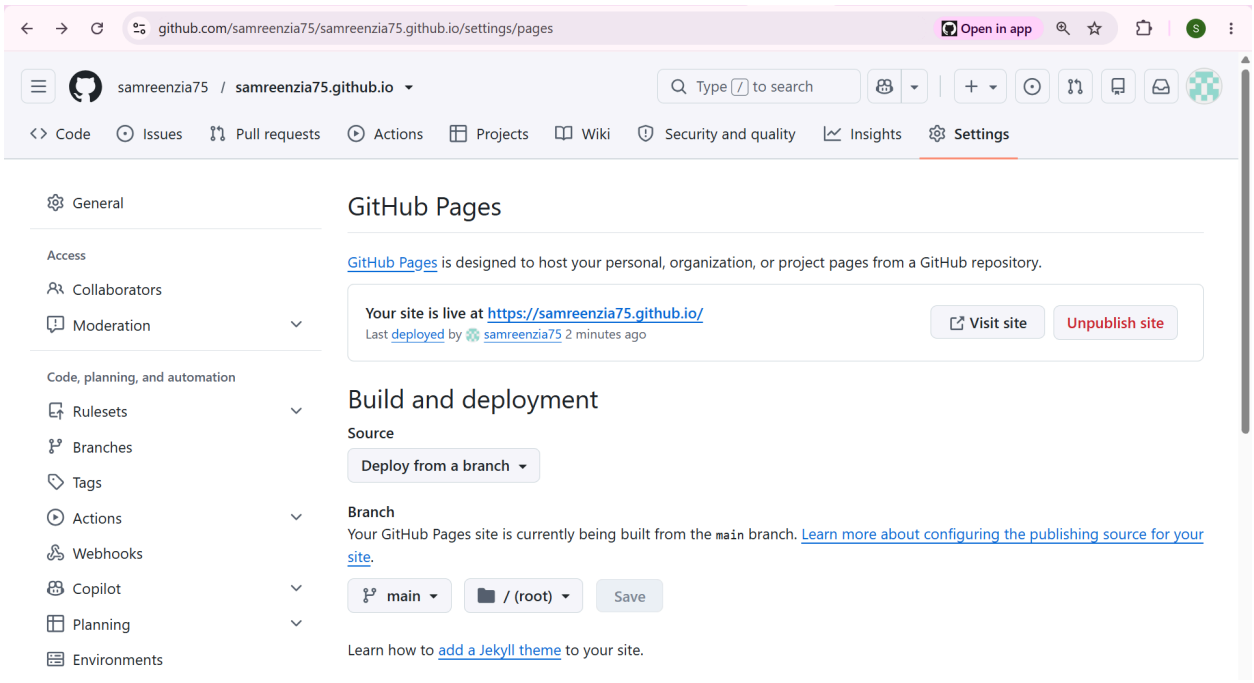
### Screenshot 1: Portfolio repository

`task4_pages_repository.png.png`



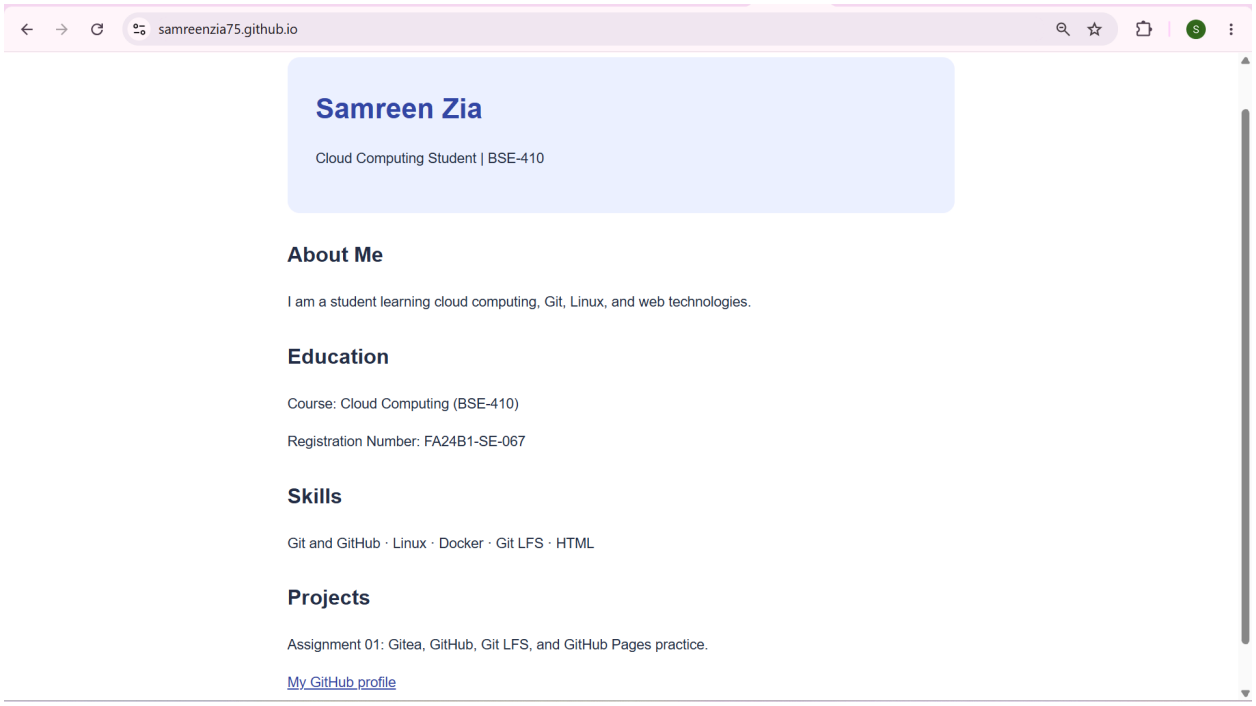

## Screenshot 2: GitHub Pages deployment settings

`task4_pages_deployment.png.png`




## Screenshot 3: Live portfolio website

`task4_portfolio_live.png.png`




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## Conclusion

This assignment provided practical experience with Git repositories, remote configuration, Gitea, GitHub, Git LFS, and GitHub Pages. The `Assignment01` repository was pushed to Gitea and GitHub, three large files were tracked with Git LFS and pushed to GitHub, and a portfolio homepage was published through GitHub Pages.